Kuldeep Singh
Innovaccer Inc.
kuldeep.singh@innovaccer.com
ORCID: 0009-0004-8476-0118

May 2, 2026

Editor-in-Chief
IEEE Transactions on Image Processing

Dear Editor-in-Chief,

I am submitting the manuscript **"A 16-Bit-Native Entropy Coding Pipeline for Lossless Compression of High-Bit-Depth Images"** for consideration as a Full Paper in *IEEE Transactions on Image Processing*.

Summary and core contribution. This paper addresses a fundamental mismatch in lossless image compression: most entropy coders are optimized for 8-bit byte streams, yet high-bit-depth images (medical scanners, scientific instruments, HDR cameras) store pixel data at 10–16 bits per sample. Applying a byte-oriented coder to 16-bit data requires splitting each pixel into a byte pair before entropy coding, losing the mutual information between the two bytes. I quantify this structural cost as 0.3–1.1 bits per symbol after spatial prediction, and present MIC—a lossless codec that eliminates it by coding 16-bit residuals as atomic symbols.

MIC combines spatial prediction, a 16-bit-native run-length representation, and an extended Finite State Entropy (FSE/ANS) coder supporting up to 65,535-symbol alphabets. This contrasts with Zstandard's FSE capped at 4,096 symbols and Huffman coding, which becomes impractical above roughly 4,096 symbols. The codec also includes a four-state interleaved decoder that exploits instruction-level parallelism (ILP), yielding a geometric-mean 68% isolated decompression speedup on ARM64 (43% on AMD64) with no compressed-size penalty.

Relevance to IEEE TIP. The manuscript makes contributions in image coding, entropy coding, and decoder algorithm design—all within the core scope of IEEE TIP. The work:

- proposes and analyzes a new entropy coding approach for large symbol alphabets arising in high-bit-depth image residual streams;
- provides an information-theoretic analysis (mutual information of byte-split residuals) that explains and predicts the compression advantage;
- evaluates a multi-state decoder design that improves throughput via instruction-level parallelism, a technique of broad relevance to image codec design.

The evaluation uses 21 de-identified high-bit-depth images (nine modalities, sizes up to 4774×3064 pixels), with fair in-process comparisons against High-Throughput JPEG 2000 (HTJ2K) and JPEG-LS, two codecs within IEEE TIP's reference space.

Suggested EDICS categories. `COD-CMP` (Compression); `COD-ENT` (Entropy Coding); `ANA-BIO` (Biomedical image analysis) as a secondary category.

Originality. This work has not been submitted to, or published in, any other journal. The manuscript is an original work; no prior conference or workshop version has been published.

Open-source reproducibility. The complete implementation, benchmark harness, and test data references are at https://github.com/pappuks/medical-image-codec. All reported results are reproducible using the commands in the repository.

Ethics. All 21 test images are from publicly available, de-identified DICOM datasets (NEMA WG-04 compsamples, NEMA 1997 CD, Clunie Breast Tomosynthesis Case 1). No human subjects research was conducted; no IRB approval is required.

Conflict of interest. The author declares no competing financial interests or personal relationships that could have influenced this work.

Suggested reviewers. Researchers active in ANS/FSE entropy coding, image coding standards (HTJ2K, JPEG-LS), or high-throughput codec design would be well placed to review this work. Authors of OpenJPH and CharLS are compared against in the paper and may have a conflict of interest at the Editor's discretion.

I confirm this is a single-author submission and that the manuscript complies with IEEE Transactions on Image Processing formatting requirements, including the 13-page initial-submission limit and the 150–250 word abstract.

Sincerely,

Kuldeep Singh
Innovaccer Inc.
kuldeep.singh@innovaccer.com
ORCID: 0009-0004-8476-0118
https://github.com/pappuks/medical-image-codec